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July 3, 2017

To Whom It May Concern

Re: 2014.09.08 90min.wav

Expert Analysis of Audio Recording

Dear Sir/Madame,

I am the founder and president of AudioPaint Ltd., a New York audio/video production company that creates and produces audio-video content for television, radio, film, and the internet, as well as for artists, publishers, recording companies, and a range of corporations.

I also am the founder and president of Legal Audio Video, which is the forensic services division of AudioPaint Ltd. In that capacity, I have been retained and have consulted on issues relating to the inspection, analysis, authentication, and enhancement of audio and video recordings by, among other entities, the United States Attorney's Offices for the Southern and Eastern Districts of New York, the Federal Bureau of Investigation, the Drug Enforcement Administration, and the Legal Aid Society in Manhattan.

I have qualified as an expert witness in the area of audio and video forensic analysis in proceedings in both state and federal court on behalf of governmental entities. Annexed to this affidavit is a copy of my professional vitae.

Conclusion:

I have had an opportunity to review "2014.09.08 90min," an audio file identified to me in an email on June 28, 2017. After completing the analysis I came to the conclusion that the recorded conversation is free from any editing or alterations. The file presents itself as fully authentic from start to finish. Critical listening techniques, waveform analysis and spectrographic analysis confirms these findings.

Details:

On June 30, 2017, I downloaded the file "2014.09.08 90min" to my computer from the website www.soundcloud.com. The original file was renamed and uploaded as a WAV file type. The following link was used for this: https://m.soundcloud.com/user-6313191/20140908-90min. The file was then opened and reviewed in numerous audio software programs.

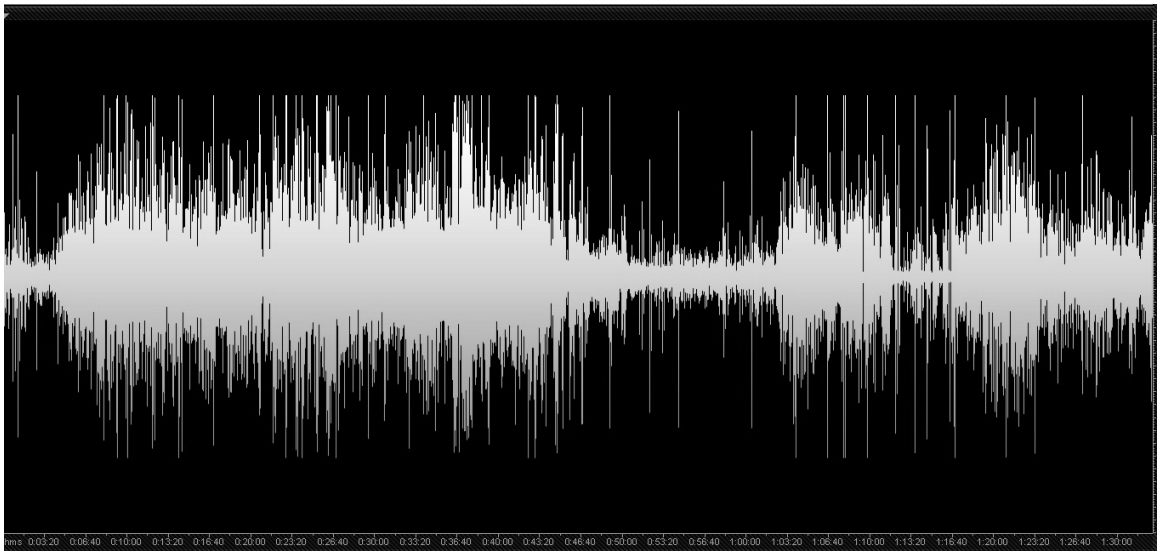
Below is a screenshot of the Sound Cloud page where file was downloaded:

The conversation appears to have taken place in a small/medium sized room between three speakers. Microphone proximity and positioning changes indicated the speakers varied in their distance from the recording device. At about 0:45 seconds of the recording, movement and handling signatures indicate the device was repositioned. Throughout the remainder of the recording the device remained stable. Transients were observed but none indicated an unnatural break in the audio. The background environment included typical office sounds such as HVAC, footsteps, doors opening and closing, etc. Street, traffic noise and outdoor environmental noise was also present.

Using critical listening techniques, I began the process of trying to locate any signatures or activity that could lead me to identify the presence of any anomalies. No anomalies were located. Spotting notes were created and details about the recording in playback time or chronological order were written.

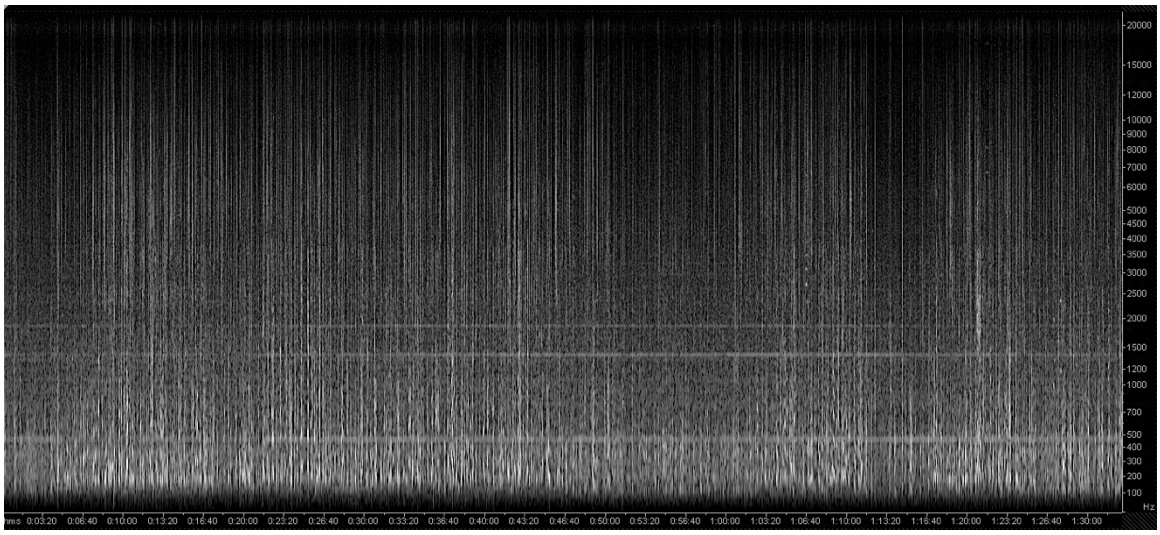
Using wave form inspection techniques, I was able to create a visual of the recording. This graphic enables me to observe time, length and gain or volume drops/peaks within the audio. There were no breaks or unnatural anomalies present.

Below is a graphic of the entire file presented as a wave form:



Using a spectrographic image I was able to locate numerous aspects of the audio including time, length, frequencies, harmonics, gain and sensitivity. This image also enables me to identify the voices and the patterns of the speakers. There were no breaks or unnatural anomalies present.

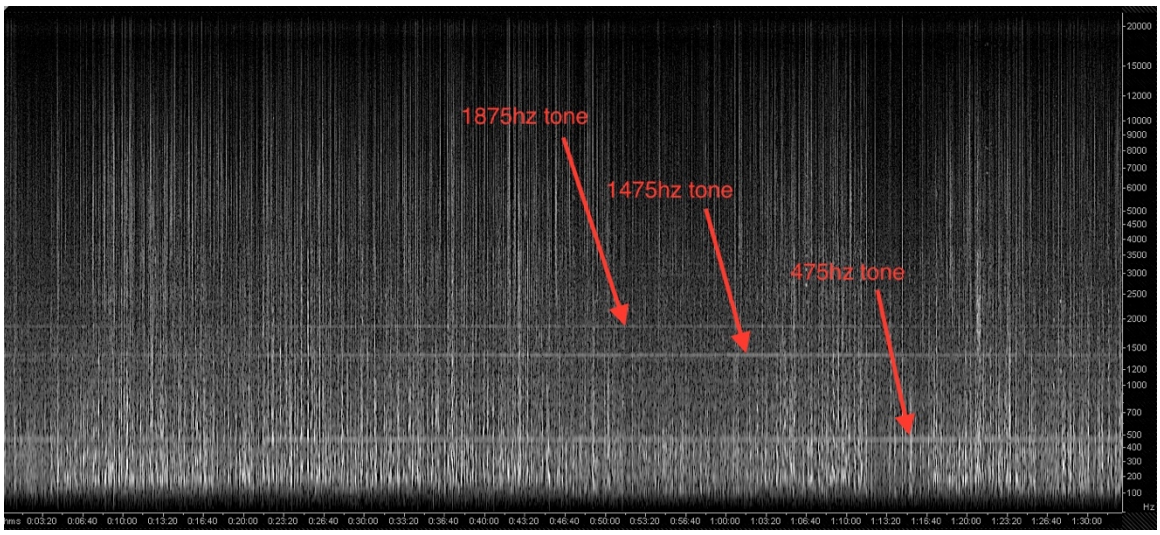
Below is a graphic of the entire file presented as a spectrographic image:



Three harmonic areas were located and extracted:

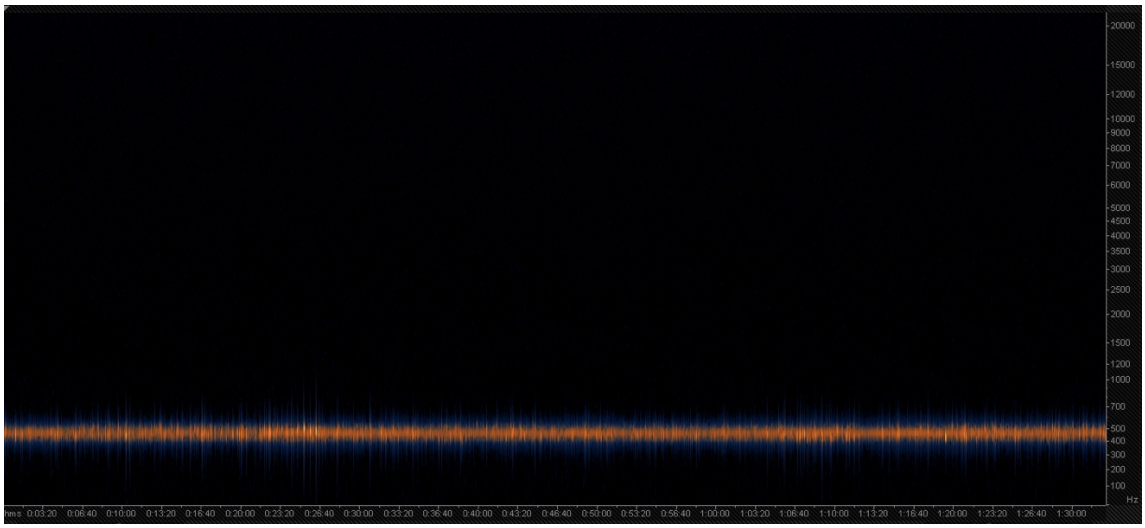
1. A 475hz harmonic tone was identified and extracted.
2. A 1400hz harmonic tone was identified and extracted.
3. A 1875hz harmonic tone was identified and extracted.

Below is a spectrographic image identifying all three tones:

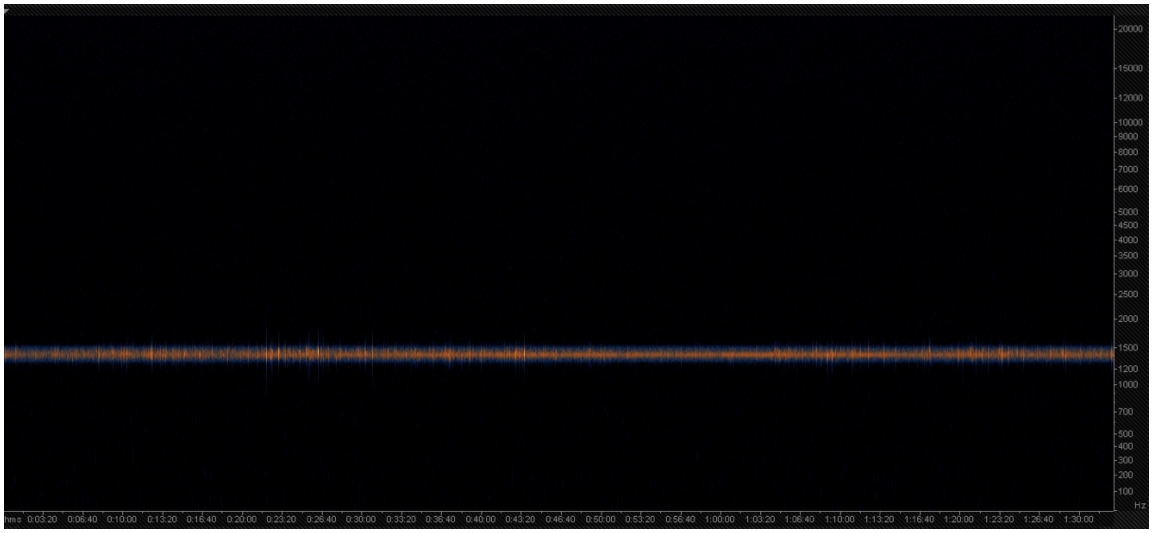


By identifying and examining the harmonic tones, I was able to locate any breaks, changes or edits by observing and reviewing their patterns. If there is a change or dropout in the tone this would indicate the presence of an unnatural shift in the environment and/or the recording. All three harmonics observed remained constant.

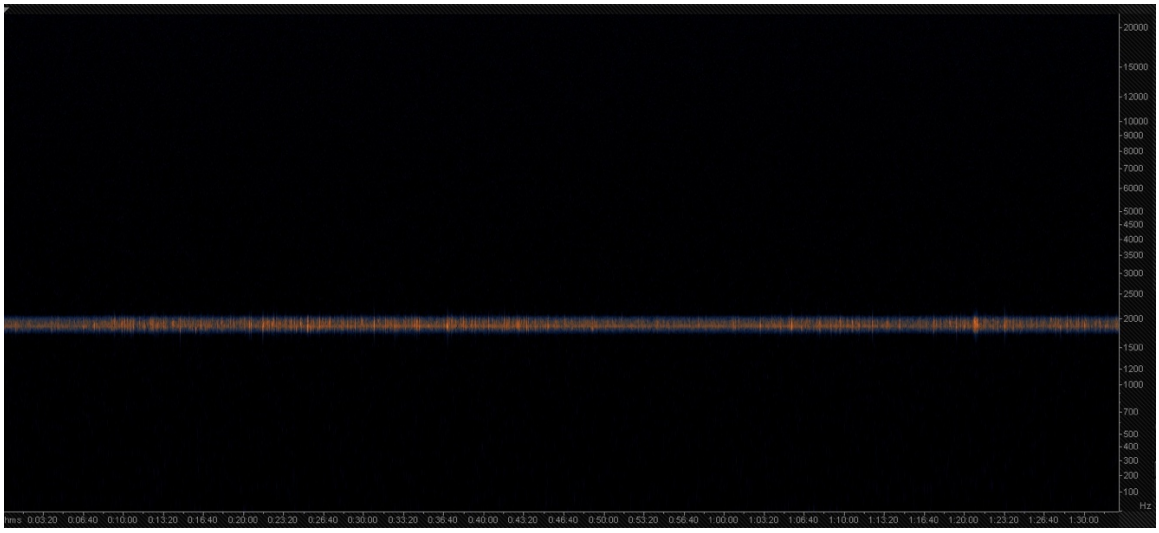
Below is a spectrographic image of the 475hz harmonic tone:



Below is a spectrographic image of the 1400hz harmonic tone:



Below is a spectrographic image of the 1875hz harmonic tone:

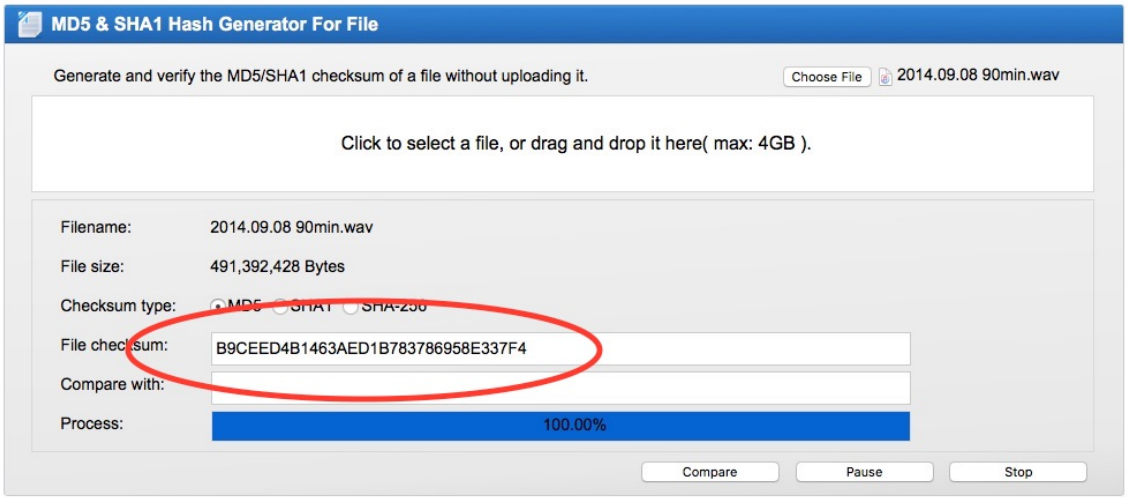


Below is a graphic of the basic file information:

```
General
Complete name                            :
Format                                   : Wave
File size                                : 469 MiB
Duration                                 : 1h 32mn
Overall bit rate mode                    : Constant
Overall bit rate                         : 706 Kbps

Audio
Format                                   : PCM
Format settings, Endianness              : Little
Format settings, Sign                    : Signed
Codec ID                                 : 1
Duration                                 : 1h 32mn
Bit rate mode                            : Constant
Bit rate                                 : 705.6 Kbps
Channel(s)                               : 1 channel
Sampling rate                            : 44.1 KHz
Bit depth                                : 16 bits
Stream size                              : 469 MiB (100%)
```

Below is the HASH signature of the file:



Missing from the analysis was the opportunity to examine the original file. Having access to this file could give me information found in the metadata that would link the file to the IPad2 Mini. Additionally, the HASH signature of the original file could be compared to the HASH on the copy file. Creating exemplar files from the original device can help to confirm certain findings that were identified in a linear analysis.

This analysis took place between June 30th and July 3rd, 2017. I was told an IPad2 Mini was used to make the recording. I did not have access to the original recording device nor the location where the recording took place. All information and findings are the result of an examination of the file provided to me. No other copy of the file was offered for this analysis.

This examination was performed using the best of my abilities as a professional audio engineer. I am confident my findings are true and accurate. No person other than myself took part in this analysis.

Sincerely,

Frank Piazza
President, Legal Audio Video